

4)

ai, aii)

$$L = \iint \tilde{p}(x) \int \delta(z - F_x(u)) \mathcal{N}(u; 0, I) du \log \frac{\tilde{p}(x) \int \delta(z - F_x(u)) \mathcal{N}(u; 0, I) du}{q(x|z) q(z)} dz dx$$

$$= \int \tilde{p}(x) \iint \delta(z - F_x(u)) \mathcal{N}(u; 0, I) \log \frac{\tilde{p}(x) \int \delta(z - F_x(u)) \mathcal{N}(u; 0, I) du}{q(x|z) q(z)} dz du dx$$

$$= \int \tilde{p}(x) \int \mathcal{N}(u; 0, I) \log \frac{\tilde{p}(x) \int \delta(F_x(u) - F_x(u')) \mathcal{N}(u'; 0, I) du'}{q(x|z) q(z)} du dx$$

$$\int \delta(F_x(u) - \underbrace{F_x(u')}_{v}) \mathcal{N}(u'; 0, I) du' = \int \delta(F_x(u) - v) \mathcal{N}(H_x(v); 0, I) \left| \det \frac{\partial F_x(u')}{\partial u'} \right|^{-1} dv$$

$$dv = du' \left| \det \frac{\partial F_x(u')}{\partial u'} \right|$$

$$= \mathcal{N}(\underbrace{H_x(F_x(u))}_u; 0, I) \left| \det \frac{\partial F_x(u')}{\partial u'} \right|^{-1} \Big|_{u'=u}$$

$$\Rightarrow L = \iint \tilde{p}(x) \mathcal{N}(u; 0, I) \log \left(\frac{\tilde{p}(x) \mathcal{N}(u; 0, I)}{q(x|F_x(u)) q(F_x(u))} \left| \det \frac{\partial F_x(u')}{\partial u'} \right|^{-1} \Big|_{u'=u} \right) du dx$$

$\rightarrow \mathcal{N}(0, I) = q(z)$

aiii)

$$\bullet v = F_x(u) \Rightarrow dv = du \left| \det \frac{\partial F_x(u)}{\partial u} \right|$$

The term $\left| \det \frac{\partial F_x(u)}{\partial u} \right|^{-1}$ comes from the transformation

from $u \rightarrow v$.

we have to scale v according to $J_{F_x}(u)$.

• By simply setting $F_x(u) = s * u$ we have:

$$p(z|x) = \int \delta(z - s * u) \mathcal{N}(u; 0, I) du = \int \delta(z - v) \mathcal{N}(s^{-1} * v; 0, I) s^{-1} * dv$$

$\Rightarrow p(z|x) = s^{-1} * \mathcal{N}(s^{-1}z; 0, I)$ we can have any $\mathcal{N}(0, \text{diag}(s^{-1}))$.

b i)

$$q(x|F_x(u)) = \mathcal{N}(x; F_x^{-1}(F_x(u)), \sigma^2 I) \\ = \mathcal{N}(x; u, \sigma^2 I)$$

$$\Rightarrow \log q(x|F_x(u)) = -\frac{d}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \|u\|_2^2$$

b ii)
$$J_{F_x}(u)_{ij} = \frac{\partial F_x(u)_i}{\partial u_j} = \frac{\partial F(\sigma u + x)_i}{\partial u_j} = \sum_k \frac{\partial F(u)_i}{\partial w_k} \frac{\partial w_k}{\partial u_j}$$

$$w = \sigma u + x \Rightarrow \frac{\partial w_k}{\partial u_j} = \sigma \delta_{kj} \Rightarrow J_{F_x}(u)_{ij} = \frac{\partial F(u)_i}{\partial w_j} \sigma$$

$$\Rightarrow \log \left| \det \frac{\partial F_x(u)}{\partial u} \right| = \log \left| \det \left(\sigma \frac{\partial F(u)}{\partial w} \right) \right| = \log \left| \sigma^d \det \frac{\partial F(u)}{\partial w} \right| \\ = d \log \sigma + \log \left| \det \frac{\partial F(u)}{\partial w} \right|$$

b iii)

$$\mathcal{L} = \int \tilde{p}(x) \log \tilde{p}(x) dx + \int \mathcal{N}(u; 0, I) \log(\mathcal{N}(u; 0, I)) du$$

$$= \mathbb{E}_{\tilde{p}(x), u \sim \mathcal{N}(0, I)} \left[\log q(x|F_x(u)) + \log \mathcal{N}(F_x(u); 0, I) + \log \left| \det \frac{\partial F_x(u)}{\partial u} \right|^{-1} \right] \\ = \mathbb{E}_{\tilde{p}(x), u \sim \mathcal{N}(0, I)} \left[\underbrace{\frac{d}{2} \log(2\pi\sigma^2)}_{\text{Const}} + \underbrace{\frac{1}{2\sigma^2} \|u\|_2^2}_{\text{Const}} + \underbrace{\frac{d}{2} \log(2\pi)}_{\text{Const}} + \frac{1}{2} \|F(\sigma u + x)\|_2^2 \right. \\ \left. - d \log \sigma - \log \left| \det \frac{\partial F(u)}{\partial w} \right| \right] + \text{Const}$$

$$\Rightarrow \mathcal{L} = \mathbb{E}_{\tilde{p}(x), u \sim \mathcal{N}(0, I)} \left[\frac{1}{2} \|F(\sigma u + x)\|_2^2 - \log \left| \det \frac{\partial F(u)}{\partial w} \right| \right] + \text{Const}$$

5)

$$a) A = \emptyset \Rightarrow \bar{A} = B \Rightarrow P(\bar{A}) = P(B) = 0.9 \times 1$$

$$\Rightarrow \forall x, y \in \mathbb{R} \quad P_\theta(x, y) < 0.9 - \epsilon \Rightarrow P_\theta(\bar{A}) = P_\theta(B) < (0.9 - \epsilon) \times 1$$

$$\Rightarrow |P_\theta(\bar{A}) - P(\bar{A})| > \epsilon$$

$$\text{Pinker's inequality: } \sqrt{\frac{1}{2} D_{KL}(P_\theta \| P)} \geq \sup_E \{ |P_\theta(E) - P(E)| \} \\ \geq P_\theta(\bar{A}) - P(\bar{A}) > \epsilon$$

$$\Rightarrow D_{KL}(P_\theta \| P) > 2\epsilon^2$$

b)

$$C = \{z \in \mathbb{R}^2; f_\theta^{-1}(z) \in A\} \quad dx = dz \left| \det \frac{\partial f_\theta(z)}{\partial z} \right|^{-1}$$

$$\text{Vol}(A) = \int_A dx = \int_C \underbrace{\left| \det \frac{\partial f_\theta(z)}{\partial z} \right|^{-1}}_{=1} dz = \int_C dz = \text{Vol}(C)$$

$$P_\theta(x) = P(z) \Rightarrow \forall z \in C \quad P(z) \geq 0.9 - \epsilon \quad \epsilon > 0$$

$$z \sim \mathcal{N}(0, I) \Rightarrow P(z) = \frac{1}{(\sqrt{2\pi})^2} \exp\left(-\frac{\|z\|_2^2}{2}\right) \geq 0.9 - \epsilon \quad \forall z \in C$$

$$\left. \begin{aligned} P_\theta(A) &\geq (0.9 - \epsilon) \text{Vol}(A) \\ P(C) &\leq \frac{1}{2\pi} \pi \left(-2 \log(2\pi(0.9 - \epsilon)) \right) \end{aligned} \right\} \Rightarrow \text{Vol}(A) \leq -\log(2\pi(0.9 - \epsilon))$$

$$c) \quad P(B) = 0.9 \times 1$$

$$P_\theta(B) = P(B) = \int_B P(z) dz \leq \frac{1}{2\pi} \int_B dz \stackrel{\rightarrow 1}{=} \frac{\text{Vol}(B)}{2\pi} = \frac{1}{2\pi}$$

$$\Rightarrow |P_\theta(B) - P(B)| = |0.9 - \frac{1}{2\pi}| > 0 \Rightarrow D_{KL}(P_\theta \| P) \geq 2(0.9 - \frac{1}{2\pi})^2 > 0$$

d)

$$D_{KL}(P_\theta \| P) \geq 2(0.9 - \frac{1}{2\pi})^2 \Rightarrow \lim_{n \rightarrow \infty} P_{\theta_n}(x) - P(x) \neq 0$$

$\Rightarrow P_{\theta_n}(x)$ can never be a universal approximator.